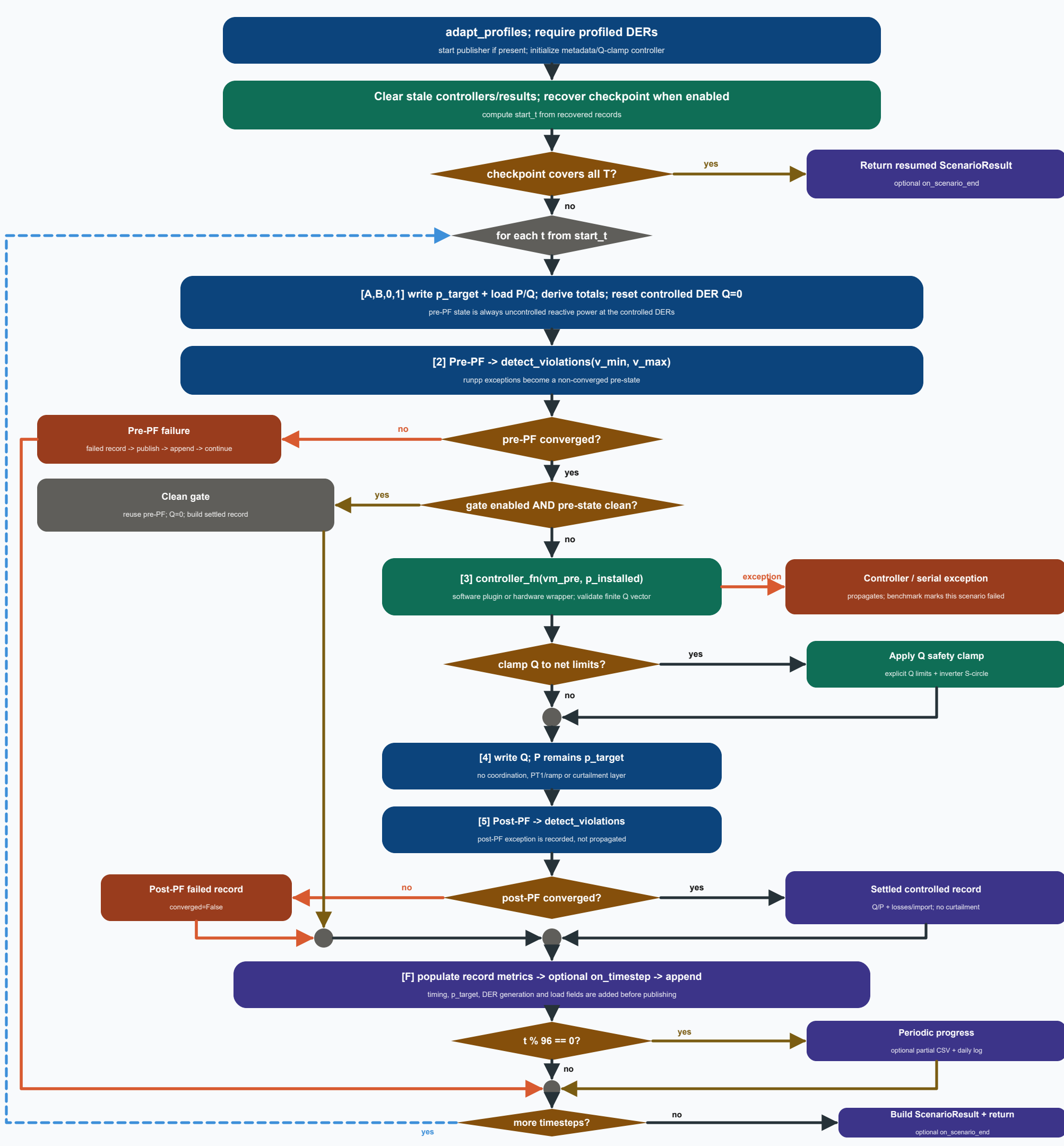




Scenario initialization and resume

The runner adapts profile tables and requires at least one profiled DER before entering the custom-control path.

A dry-run VoltVarController is created only to reproduce built-in DER ordering, bus mapping, installed-P resolution and Q limits.

No serial interface is opened by that metadata controller. Hardware serial ownership remains in the controller_fn wrapper.

Existing pandapower controllers/results are cleared so this scenario starts from an isolated simulation state.

When checkpointing is available, prior TimestepRecords are recovered and start_t advances to the first missing timestep.

A complete checkpoint is converted directly into ScenarioResult and returned without repeating any power-flow timestep.

Per-timestep control sequence

Each timestep writes the current active-power/load profiles and enforces Q=0 before the uncontrolled pre-power-flow snapshot.

If the pre-PF fails, a non-converged record is published/appended immediately and the controller is not evaluated.

With gate_clean_timesteps enabled, a clean pre-state is already the settled state, so Q remains zero and no second PF is run.

Otherwise controller_fn receives pre-control DER-bus voltages plus installed/rated active power in the same DER order.

The return vector is validated before optional clamping to explicit Q limits and the inverter apparent-power capability circle.

The accepted Q vector is written directly. Active power remains p_target because this custom path has no dynamics/ramp layer.

A second PF evaluates the controlled state. Its non-convergence is recorded locally so later timesteps can still run.

Residual voltage/thermal violations remain in the record; the custom path does not perform active-power curtailment.

Publishing, progress and hardware execution

Normal, clean-gate and post-PF-failure records receive timing/profile metrics before optional on_timestep publishing and append.

Every 96th normal/clean timestep may rewrite the partial benchmark CSV and emits the daily custom-controller progress log.

The pre-PF-failure continue bypasses that periodic branch and goes directly to the next timestep.

When controller_fn is software, the imported Python algorithm computes Q locally from vm_pu and installed P.

When controller_fn is HardwareControllerFn, its first call opens/configures Arduino and each call exchanges V:/Q: setpoints.

Controller-function errors, output-validation errors and serial errors propagate out of this scenario; benchmark_runner isolates the failure.

After the final timestep, all records are aggregated into ScenarioResult and on_scenario_end is emitted when a publisher is attached.